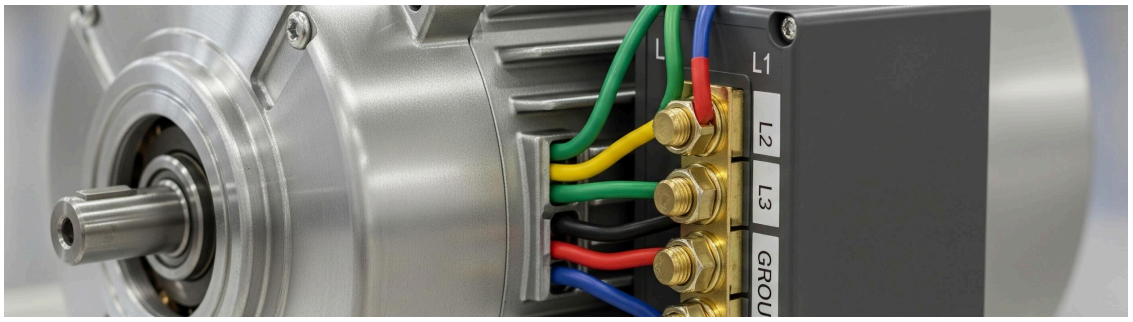

Electrical Machine Data Sheet Template

Machine Name/Type : DC Shunt motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

Parameter	Value	Unit
Power/Output	5	HP
Voltage	220	V
Current (Rated)	19	A
Frequency	NA	NA
Speed	1500	RPM
Field current	1	A

Precautions

I. Ensure the **supply voltage is 220 V DC** before switching ON the motor.

II. **Never run the motor without field excitation**. Maintain rated field current of **1 A** at all times.

III. Start the motor **only with a proper starter** to limit excessive starting current.

IV. Verify **correct and tight terminal connections** as per the circuit diagram before operation.

V. Do not allow the **armature current to exceed 19 A** during loading.

VI. Apply load **gradually** and avoid sudden load changes.

VII. Maintain proper **earthing of the motor body** to prevent electric shock.

VIII. Do not touch rotating parts while the motor is running.

IX. Monitor **speed (≈1500 RPM), noise, and temperature** continuously during operation.

X. Switch OFF the supply immediately in case of **abnormal sound, vibration, or overheating**.

Working Principle of DC Shunt Motor

A DC shunt motor operates on the principle that **a current-carrying conductor placed in a magnetic field experiences a force**.

The field winding is connected in parallel with the armature, producing **nearly constant flux**. Interaction between armature current and magnetic field produces torque, causing the motor to rotate. Hence, the motor runs at **almost constant speed**.

Working Characteristics of DC Shunt Motor

I. The **speed remains nearly constant** from no-load to full-load conditions.

II. Torque is **directly proportional to armature current**.

III. Speed decreases slightly with increase in load due to internal losses.

IV. The motor provides **good speed regulation and stable operation**.

Applications

1. Lathe machines
2. Drilling machines
3. Milling machines
4. Fans and blowers
5. Centrifugal pumps
6. Conveyors
7. Printing presses
8. Textile machinery
9. Machine tools
10. Laboratory testing equipment